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Research interests Power system optimization; offshore wind farm collector and transmission systems; distribution network planning and transportation electrification.

EDUCATION

Tsinghua University

Ph.D. in Electrical Engineering | Advisor: Prof. Xinwei Shen

Sep. 2024 – Present

Tsinghua University

Research Assistant in Electrical Engineering | Advisor: Prof. Xinwei Shen

Sep. 2023 – Aug. 2024

National University of Singapore

M.Sc. in Electrical Engineering | Advisor: Prof. Jimmy C.-H. Peng

Sep. 2022 – Jun. 2023

Chongqing University

B.Eng. in Electrical Engineering and Automation

Sep. 2018 – Jun. 2022

PUBLICATIONS

Journal Papers

- [J1] **W. Gao**, W. Xu, Y. Du, Y. Wang, and X. Shen, “Life-Cycle Planning of Collector System for Deep-Sea Multi-Spatial Wind-PV-Tidal Farm.” *CSEE Journal of Power and Energy Systems*, 2026.
- [J2] Y. Wang¹, **W. Gao**¹, and X. Shen, “Integrated Planning of Urban Distribution Network and Multi-Charging Infrastructures with Smart Transportation System.” *IEEE Transactions on Transportation Electrification*, 2026. (¹ Co-first author).
- [J3] **W. Gao**, Y. Wang, W. Chen, and X. Shen, “Distributionally Robust Joint Planning of Coastal Distribution Network and PV-Storage-EV Stations.” *Applied Energy*. (Second-round review).
- [J4] **W. Gao**, Y. Wang, Y. Luo, and X. Shen, “Coordination-Aware Two-Stage Robust Planning for Multi-Stakeholder Power System Optimization.” *IEEE Transactions on Industrial Informatics*. (First-round review).
- [J5] **W. Gao**, X. Shen, X. Ding, X. Ji, H. Li, K. Wang, and W. Yang, “Optimal Planning of Power Collection System for Offshore Wind Farm Based on Large-scale Mixed-integer Programming.” *Automation of Electric Power Systems*, vol. 50, no. 1, pp. 168–177, 2026 (in Chinese).
- [J6] **W. Gao**, Q. Sun, Q. Ran, P. Jian, and W. Chen, “A New High-Voltage SBD Device Structure with an Improved Buried Buffer Doped Layer.” *Microelectronics*, vol. 51, no. 1, pp. 116–120, 2021 (in Chinese).
- [J7] H. Yang, X. Shen, **W. Gao**, Y. Du, M. Shi, Q. Li, W. Wei, and Y. Chen, “Optimal Planning of Offshore Wind Farm Collector System Considering Intensive Sea Use.” *IEEE Transactions on Sustainable Energy*, 2026.
- [J8] H. Yang, X. Shen, **W. Gao**, X. Ding, Y. Du, and Q. Li, “Optimal Planning for Offshore Wind Farm Electrical Collector System Considering N-1 Criterion.” *IEEE Transactions on Sustainable Energy*, 2025.
- [J9] Y. Du, X. Shen, Y. Wang, B. Zhao, Y. Luo, and **W. Gao**, “Optimal Sizing of Isolated Electricity-Hydrogen Offshore Oilfield Integrated Energy System.” *CSEE Journal of Power and Energy Systems*, 2026.

PUBLICATIONS

Conference Papers

[C1] **W. Gao**, X. Shen, and X. Ding, “Electrical Collector System Planning for Co-Located Offshore Wind and Tidal Current Farm Considering Temporal Variability.” *2025 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, 2025.

[C2] **W. Gao** and H. Li, “Cost of Charging Electric Vehicles in a City.” *2022 IEEE PES Innovative Smart Grid Technologies-Asia (ISGT Asia)*, pp. 490–494, 2022.

[C3] C. Xu, **W. Gao**, and X. Shen, “Co-Optimization of Substation and Converter Station Locations for Large-scale Offshore Wind Farm Collection and Transmission Systems.” *2026 IEEE 10th Conference on Energy Internet and Energy System Integration (EI2)*, 2026.

[C4] J. Li, **W. Gao**, and X. Shen, “Co-Planning of Offshore Wind Farm Collection and Transmission System with Multi-Substation and Obstacle Zone.” *2025 IEEE 9th Conference on Energy Internet and Energy System Integration (EI2)*, 2025.

PROJECTS

National Natural Science Foundation of China Grant 52477102 <i>Integrated Optimization of Wind Farm Layout and Collector System Planning for Deep-Sea Offshore Wind Farms</i>	Jan. 2025 – Present
Guangdong Basic and Applied Basic Research Foundation Grant 2023A1515240055 <i>Reliability-Oriented Optimal Planning of Offshore Wind Collector Systems Considering Seabed Microenvironments</i>	Oct. 2023 – Present
Guangdong Basic and Applied Basic Research Foundation Grant 2022A1515240019 <i>Reliability-Oriented Optimal Planning of Multi-Substation Collector Systems for Offshore Wind Farms</i>	Oct. 2022 – Oct. 2025
Excellent Youth Basic Research Fund of Shenzhen Grant RCYX20231211090430053 <i>Complex Combinatorial Optimization Theory and Applications for Offshore Wind Power Planning, Operation and Maintenance</i>	Jun. 2024 – Present
CNPC Southern Petroleum Exploration and Development Company Scientific Research and Technology Development Project <i>Coordinated Optimization and Energy Efficiency Enhancement Technologies for Offshore Wind Power</i>	Apr. 2026 – Present

LEADERSHIP

Tsinghua University <i>Secretary and Class Representative, Shendian Graduate Class 42</i> Coordinated branch affairs, membership administration, records and student activities.	Aug. 2024 – Aug. 2026
Tsinghua University <i>Teaching Assistant, Advanced Power Network Analysis</i> Supported teaching, course-material preparation and course administration.	Sep. 2025 – Jan. 2026
National University of Singapore Suzhou Research Institute <i>Class Representative</i> Served as a student liaison and helped organize welcome and New Year events.	Sep. 2021 – Jun. 2022
Chongqing University <i>Secretary, Class 10, 2018 Cohort</i> Organized branch activities and supported membership records and collective-award applications.	Sep. 2020 – Jun. 2022
Chongqing University <i>Class Representative, Class 11, 2018 Cohort</i> Supported class administration, student communication and group activities.	Sep. 2018 – Apr. 2019

SKILLS

Programming MATLAB, Python, C++
Software Microsoft Office, LaTeX, ArcGIS, Photoshop
Languages Chinese (native); English (IELTS, CET-6; English-medium study and research at NUS)